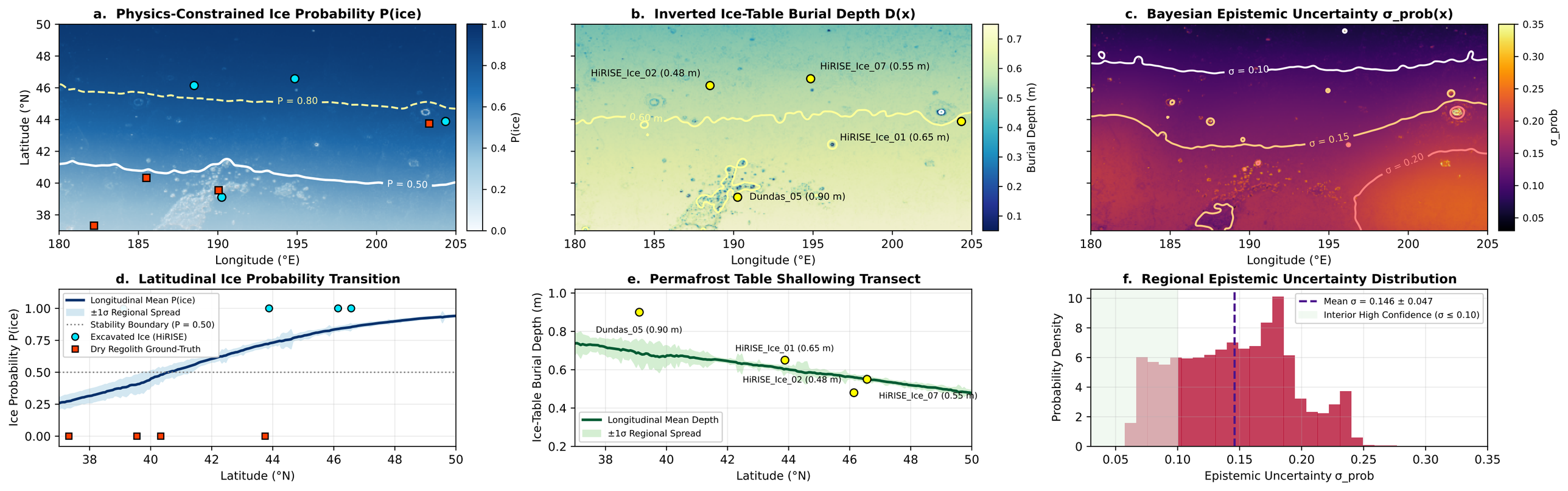

a. Physics-Constrained Ice Probability P(ice)
P = 0.80
P = 0.50
Latitude (°N)
Longitude (°E)
P(ice)
b. Inverted Ice-Table Burial Depth D(x)
HiRISE_Ice_02 (0.48 m)
HiRISE_Ice_07 (0.55 m)
0.60 m
HiRISE_Ice_01 (0.65 m)
Dundas_05 (0.90 m)
Burial Depth (m)
c. Bayesian Epistemic Uncertainty σ_prob(x)
σ = 0.10
σ = 0.15
σ = 0.20
σ_prob
d. Latitudinal Ice Probability Transition
Longitudinal Mean P(ice)
±1σ Regional Spread
Stability Boundary (P = 0.50)
Excavated Ice (HiRISE)
Dry Regolith Ground-Truth
Ice Probability P(ice)
Latitude (°N)
e. Permafrost Table Shallowing Transect
Dundas_05 (0.90 m)
HiRISE_Ice_01 (0.65 m)
HiRISE_Ice_02 (0.48 m)
HiRISE_Ice_07 (0.55 m)
Longitudinal Mean Depth
±1σ Regional Spread
Ice-Table Burial Depth (m)
Latitude (°N)
f. Regional Epistemic Uncertainty Distribution
Mean σ = 0.146 ± 0.047
Interior High Confidence (σ ≤ 0.10)
Probability Density
Epistemic Uncertainty σ_prob